ECE Department, NIT Patna (Bihta Campus)
Network Theory (EC031303) Mid Semester Exam (2 Hrs)

Attempt *all the* questions.

1. Determine the current i_1 in the circuit of Fig. 1 using nodal analysis method and graph theory concepts. (CO 1, BL 3, **3M**).
2. In the following circuit of Fig. 2, determine the voltagesV_2 and V_3 using cut-set analysis. Select the circuit elements (1), (2) and (3) in the tree (CO 1, BL 3, **3M**).
3. The linear oriented graph is given in Fig. 3, considering a tree, mark all the fundamental cut-sets and form the cut-set matrix. (CO 1, BL 3, **2M**).
4. Determine the current i_2 as labelled in the circuit of Fig. 4, with the assistance of nodal analysis (CO 2, BL 3, **2M**).

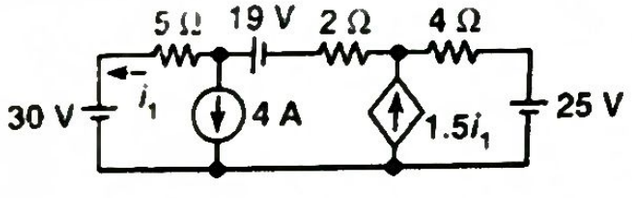


Fig. 1

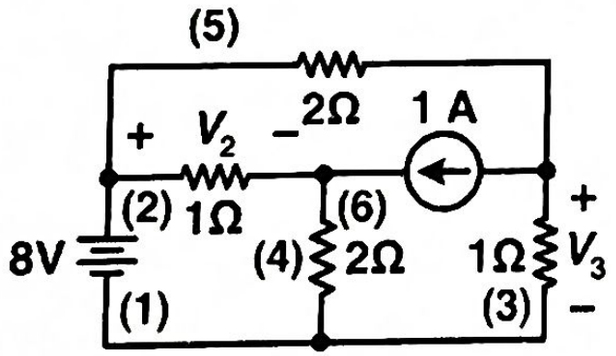


Fig. 2

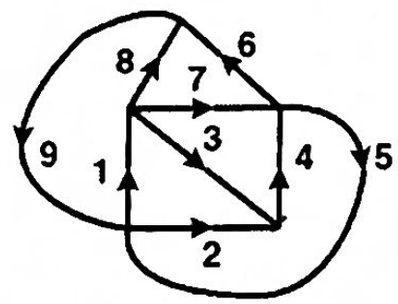


Fig. 3

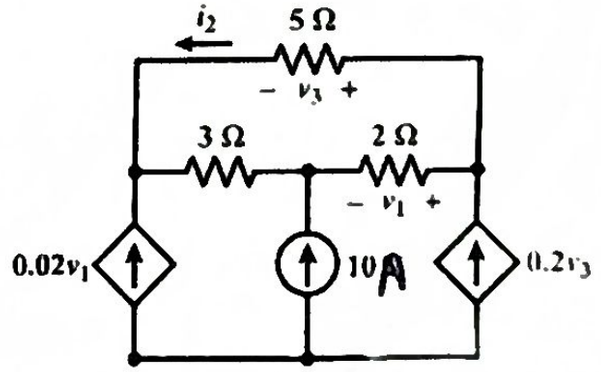


Fig. 4

5. Find the impedance matrix of the network shown in Fig. 5. (CO 2, BL 5, 2M).
6. The z-parameters of the two-port network N in Fig. 6 are Z_{11} 4s, Z_{12} Z_{21} 3s, Z_{22} 9s. (a) Replace N by its T-equivalent, (b) Use part (a) to find the input current I_1 for $V_s = cos1000t$. (CO 2, BL 5, 3M).
7. For the network shown in Fig. 7, determine the *y* parameters. (CO 2, BL 5, 2M).

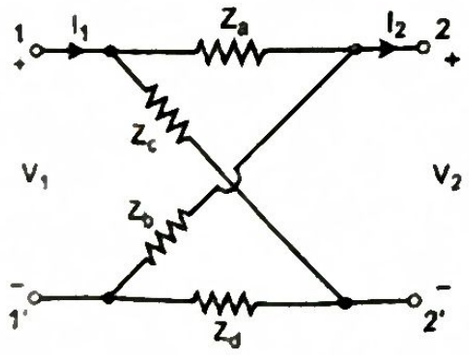


Fig. 5

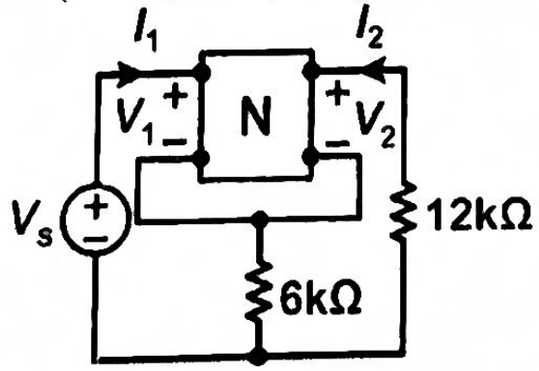


Fig. 6

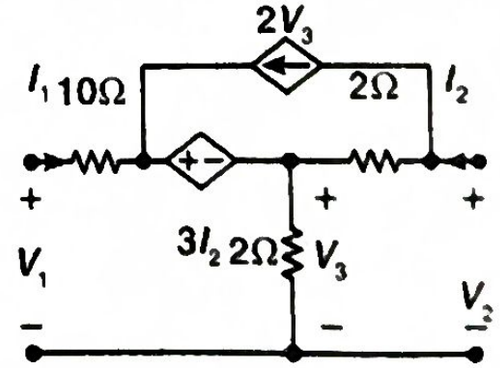


Fig. 7

8. A series circuit consisting of a coil and a capacitor is excited by a sinusoidal voltage source of E volts and variable frequency. The resonant frequency of the circuit is f_o and quality factor of the circuit is Q. Calculate the frequency at which the ratio of capacitor voltage to the source voltage is maximum and the maximum value of this ratio. (CO 2, BL 4, **3M**).
9. Show that the sum of energy stored by the inductor and the capacitor connected in series at resonance at any instant is constant and is given by LI^2. (CO 2, BL 4, **2M**).
10. For the circuit shown in Fig. 8, determine (*a*) $v_C(0^-)$; (*b*) $v_C(0^+)$; (*c*) the circuit time constant; (*d*) v_C at 3 *ms*). (CO 2, BL 3, **2M**).
11. The circuit depicted in Fig. 9 contains two independent sources, one of which is only active for $t > 0$. (*a*) Obtain an expression for $i_L(t)$ valid for all t; (*b*) calculate $i_L(t)$ at $t = 10\ \mu s$, 20 μs, and 50 μs. (CO 2, BL 3, **2M**).
12. With regard to the circuit represented in Fig. 10, determine (*a*) $i_C(0^-)$; (*b*) $i_L(0^-)$; (*c*) $i_R(0^-)$; (*d*) $v_C(0^-)$; (*e*) $i_C(0^+)$; (*f*) $i_L(0^+)$; (*g*) $i_R(0^+)$; (*h*) $v_C(0^+)$. (CO 2, BL 3, **2M**).
13. Obtain an equation for v_C as labelled in the circuit of Fig. 11 valid for all $t > 0$. (CO 2, BL 3, **2M**).

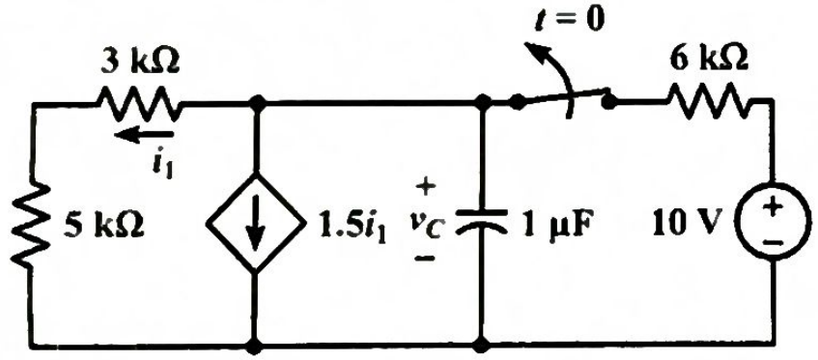


Fig. 8

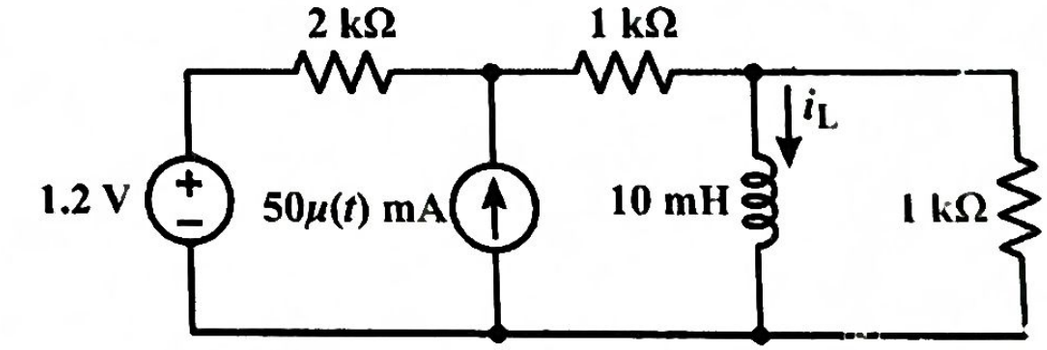


Fig. 9

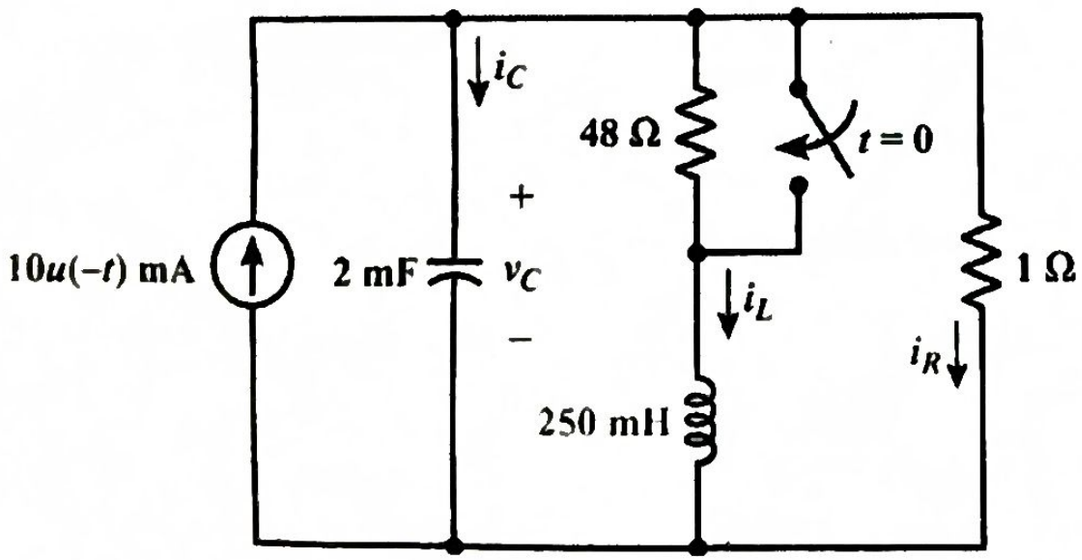


Fig. 10

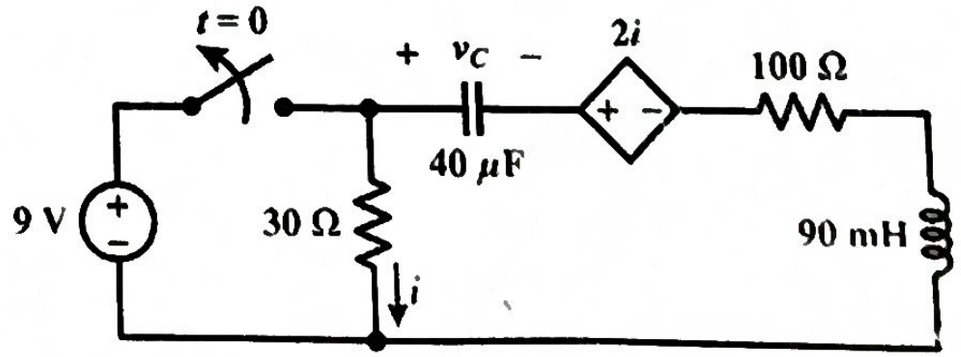


Fig. 11